Early leafout extends calendar but not thermal growing seasons

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Abstract

Recent results suggest earlier leafout does not always increase plant growth, but have struggled to explain why. Here, using rarely available plant-level phenology from a common garden of 13 woody species, we show that earlier leafout leads to earlier growth cessation (budset), and thus longer but cooler growing seasons, especially for early-leafing species. These results challenge fundamental assumptions about calendar versus thermal growing seasons with implications for carbon models.